

DATA CLEANING & PREPARATION REPORT

Project Title: Data Cleaning & Preparation

Internship: DecodeLabs Data Analytics Internship

Tool Used: Microsoft Excel

Dataset: E-commerce Orders Dataset

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OBJECTIVE:

The objective of this project was to clean and prepare a raw e-commerce dataset by identifying and handling missing values, removing duplicate records, standardizing data formats, and validating data quality. The goal was to ensure that the dataset is accurate, consistent, and ready for further analysis.

DATA CLEANING PROCESS:

1. Missing Value Analysis:

The dataset was examined for missing values across all columns.

Findings:

- Missing values were found in the CouponCode column.
- Other columns did not contain missing values.

Action Taken:

- Blank CouponCode values were replaced with "No Coupon Used" to indicate that no promotional code was applied during the purchase.

2. Duplicate Record Check

The dataset was checked for duplicate OrderIDs using Excel's Remove Duplicates feature and validation techniques.

Result:

- No duplicate OrderIDs were found.
- Dataset maintained unique order records.

3. Date Format Standardization

The Date column was reviewed for formatting consistency.

Issue Identified:

- Date values were not standardized.

Action Taken:

- All dates were converted into a consistent format (YYYY-MM-DD).

4. Data Validation

The following validation checks were performed:

Quantity Validation

- Verified that quantity values were positive.
- No invalid quantities were found.

Unit Price Validation

- Verified that all unit prices were positive numeric values.
- No invalid prices were found.

Total Price Verification

- Verified that TotalPrice equals Quantity × UnitPrice by making a column 'CalculatedPrice' and 'CompareValues'.
- Data consistency was maintained.

Tracking Number Validation

- Verified consistency in tracking number formatting.

SUMMARY OF CLEANING ACTIVITIES

At Last, I created a new sheet for the Validation.

CONCLUSION:

The dataset was successfully cleaned and validated using Microsoft Excel. Missing values in the CouponCode column were handled appropriately, date formats were standardized, and data quality checks confirmed the integrity of the dataset. The cleaned dataset is now suitable for further analysis and reporting.

